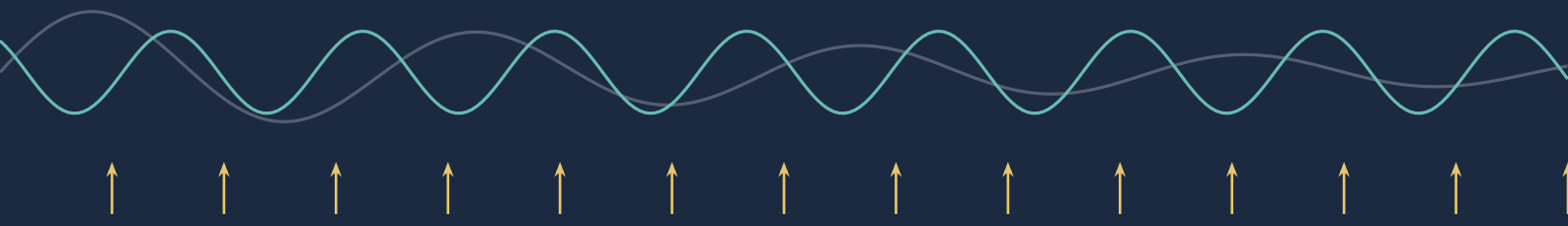
BUET • L-2/T-II • B.Sc. Engineering Examinations 2024–2025

CSE 219

Signals and Linear Systems

Term Final 2024–25 • Complete Worked Solutions



8	31	210	34
questions	parts solved	full marks	figures & graphs

Question paper dated 02/11/2026, Sub: CSE 219 (Signals and Linear Systems), Full Marks 210, Time 3 hours. Section A (Questions 1–4): sampling, DFT/FFT, z- and Laplace transforms built around real devices. Section B (Questions 5–8): signals, convolution, Fourier series and Fourier transform.

Every number in this book was re-derived independently and checked numerically. Wherever the question leaves a choice open, the choice is stated in a note box.

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How to read this book

The setting	The story or background shared by all parts of a question, copied from the paper.
Question card	Each sub-question with its marks badge, exactly as set.
Solution	Step-by-step working: the idea first, then the calculation, then a check.
Key idea	The one principle the examiner is testing — worth remembering for other papers.
Note	Where the question allows a choice (a sketch parameter, a convention), the choice is stated here.
Answer	The final result, boxed, ready to copy into a script.

> **KEY IDEA** Section A questions are long "application" stories. Read the setting once, translate every sentence into a signal-and-system statement (input, system, output, sampling rate, transform), and the sub-questions become routine.

Conventions used throughout

- $\omega = 2\pi f$; $W_N = e^{-j2\pi/N}$; $X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}$.
- Fourier series $x(t) = \sum_k a_k e^{jk\omega_0 t}$, $a_k = \frac{1}{T}\int_T x(t) e^{-jk\omega_0 t}\,\mathrm{d}t$.
- Fourier transform $X(j\omega) = \int x(t) e^{-j\omega t}\,\mathrm{d}t$; Laplace $X(s) = \int x(t) e^{-st}\,\mathrm{d}t$; z-transform $X(z) = \sum x[n] z^{-n}$.

Section A

Questions 1–4

Sampling • DFT & FFT • z-transform • Laplace transform

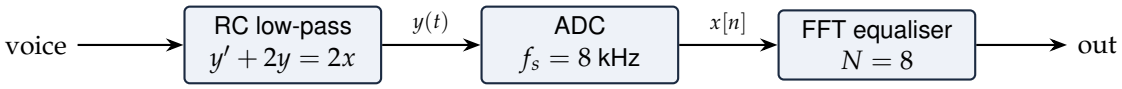
01

The USB Microphone: Filter, Sampler and FFT Equaliser

QUESTION 1 • COMPULSORY • 35 MARKS

THE SETTING

A USB microphone has three stages: an analogue circuit containing an RC (resistor–capacitor) low-pass filter; an analogue-to-digital converter (ADC) that samples at $f_s = 8$ kHz; and a digital equaliser in software.



Q1(a) Impulse and step response — MARKS 11

The analogue filter obeys

$$\frac{dy(t)}{dt} + 2y(t) = 2x(t).$$

Solve for the impulse response $h(t)$ of the filter and for its output when the input is the unit step $u(t)$.

SOLUTION

Step 1 — Transfer function. Take the Laplace transform with zero initial conditions (the filter starts at rest):

$$sY(s) + 2Y(s) = 2X(s) \quad \Longrightarrow \quad H(s) = \frac{Y(s)}{X(s)} = \frac{2}{s+2}.$$

Step 2 — Impulse response. For $x(t) = \delta(t)$, $X(s) = 1$, so $Y(s) = H(s)$. Using $e^{-at}u(t) \longleftrightarrow \frac{1}{s+a}$:

$$\boxed{h(t) = 2e^{-2t}u(t)}$$

Time-domain check. For $t > 0$ the input is zero, so $h' + 2h = 0 \Rightarrow h = Ce^{-2t}$. Integrating the equation from 0^- to 0^+: $h(0^+) - h(0^-) = 2\int \delta = 2$, so $C = 2$. ✓

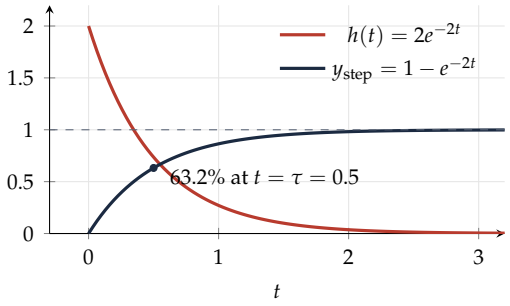
Step 3 — Step response. $X(s) = \frac{1}{s}$, so

$$Y(s) = \frac{2}{s(s+2)} = \frac{A}{s} + \frac{B}{s+2}, \qquad A = \left[\frac{2}{s+2}\right]_{s=0} = 1, \quad B = \left[\frac{2}{s}\right]_{s=-2} = -1.$$

$$\boxed{y_{\text{step}}(t) = \left(1 - e^{-2t}\right)u(t)}$$

Equivalently $y_{\text{step}}(t) = \int_0^t h(\tau)\,\mathrm{d}\tau = \int_0^t 2e^{-2\tau}\,\mathrm{d}\tau = 1 - e^{-2t}$.

Step 4 — Interpretation. The time constant is $\tau = RC = \frac{1}{2}$ (in the units of the question), the d.c. gain is $H(0) = 1$ (a slow input passes unchanged) and the -3 dB cutoff is $\omega_c = 2$ rad/s: $|H(j\omega)| = \frac{2}{\sqrt{4+\omega^2}}$ falls to $\frac{1}{\sqrt{2}}$ at $\omega = 2$.



Answer. $H(s) = \dfrac{2}{s+2}$, $h(t) = 2e^{-2t}u(t)$; step response $y(t) = (1 - e^{-2t})u(t)$, rising to 1 with time constant 0.5.

Q1(b) Aliasing of the noise tone — MARKS 11

The microphone picks up a voice, band-limited to 3.4 kHz, together with a 7 kHz noise tone from a laptop power supply, which the analogue filter does not remove. **Explain**, with a sketch of the spectrum of the sampled signal showing the voice and the noise tone, whether aliasing has occurred and how it can be removed.

SOLUTION

Step 1 — What sampling does to a spectrum. Sampling at f_s repeats the analogue spectrum at every multiple of f_s:

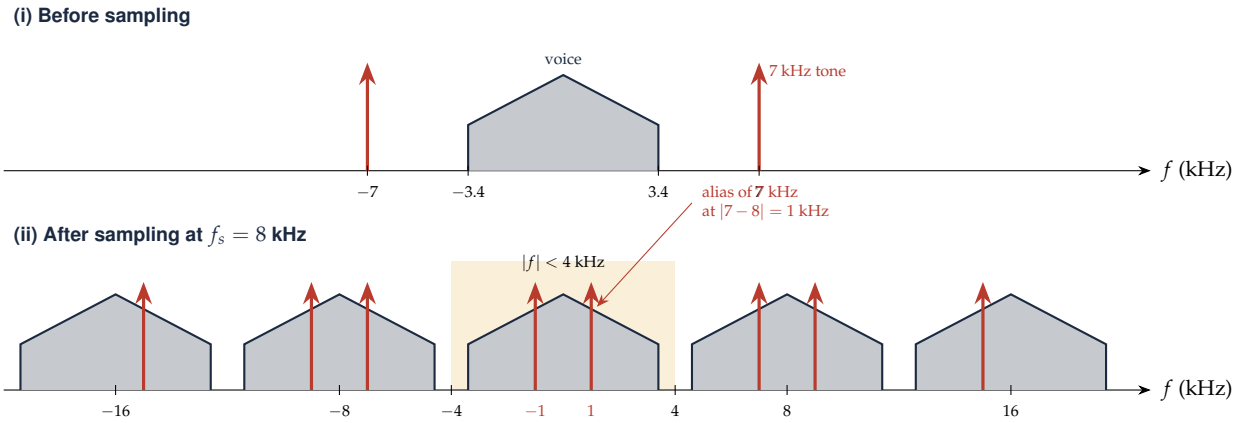
$$X_s(f) = f_s \sum_{k=-\infty}^{\infty} X(f - kf_s), \qquad f_s = 8\ \text{kHz}, \qquad \text{usable band } |f| < \tfrac{f_s}{2} = 4\ \text{kHz}.$$

Step 2 — The voice. The voice occupies $|f| \le 3.4$ kHz < 4 kHz. Its copies sit at $0, \pm 8, \pm 16, \ldots$ kHz and are separated by gaps of $8 - 2(3.4) = 1.2$ kHz. *No aliasing for the voice.*

Step 3 — The 7 kHz tone. 7 kHz > 4 kHz, so it violates the Nyquist condition. Its images lie at $\pm 7 + 8k$ kHz:

$$\ldots,\ -9,\ \underbrace{-1,\ +1}_{\text{inside } |f|<4},\ 9,\ 15,\ \ldots \text{ kHz}.$$

The copy shifted by $-f_s$ lands at $7 - 8 = -1$ kHz, and its mirror at $+1$ kHz. So after sampling, the tone *looks exactly like a 1 kHz tone* — right in the middle of the voice band.



Step 4 — Has aliasing occurred? Yes, for the noise tone. The voice is reproduced faithfully, but the 7 kHz tone has folded to 1 kHz and is now indistinguishable from a genuine 1 kHz voice component: any sample sequence produced by $\cos(2\pi \cdot 7000\, n/8000)$ is identical to that of $\cos(2\pi \cdot 1000\, n/8000)$, since $\cos\left(2\pi\frac{7}{8}n\right) = \cos\left(2\pi n - 2\pi\frac{1}{8}n\right) = \cos\left(2\pi\frac{1}{8}n\right)$.

Step 5 — How to remove it.

- **Before the ADC (the correct cure):** an *anti-aliasing* analogue low-pass filter with its passband up to 3.4 kHz and strong attenuation by $f_s/2 = 4$ kHz (certainly at 7 kHz). The first-order RC stage clearly is not enough; a sharper, higher-order filter (or a higher cutoff ratio) is needed.
- **Alternatively, raise the sampling rate** above $2 \times 7 = 14$ kHz (e.g. 16 kHz): the tone then stays at 7 kHz, outside the voice band, and a digital low-pass filter at 3.4 kHz removes it before decimating back to 8 kHz.
- **After sampling** it can only be *patched*: a narrow digital notch at 1 kHz removes the alias but also removes the genuine voice content at 1 kHz. Aliasing cannot be undone, because the samples no longer carry the information that the tone was at 7 kHz.

> **KEY IDEA** A frequency f_0 sampled at f_s appears at $|f_0 - k f_s|$ for the integer k that brings it into $[0, f_s/2]$. Filter *before* you sample.

Answer. The voice (≤ 3.4 kHz < 4 kHz) is not aliased, but the 7 kHz tone aliases to $|7 - 8| = 1$ kHz, inside the voice band. It must be removed by an anti-aliasing low-pass filter (cutoff $\approx$ 3.4–4 kHz) before the ADC, or by sampling faster than 14 kHz and filtering digitally.

Q1(c) 8-point radix-2 DIT FFT and the equaliser bins — **MARKS 13**

The equaliser works on blocks of $N = 8$ samples: it computes their DFT with the radix-2 decimation-in-time FFT, scales the bins it wants louder, and sends the result back through the inverse FFT. **Examine**, by computing the 8-point radix-2 decimation-in-time FFT of the test block

$$x[n] = \{1, 2, 1, 0, 1, 2, 1, 0\}$$

with its butterfly diagram, which bins the equaliser must scale to boost the vocal range, 300 Hz to 3 kHz.

SOLUTION

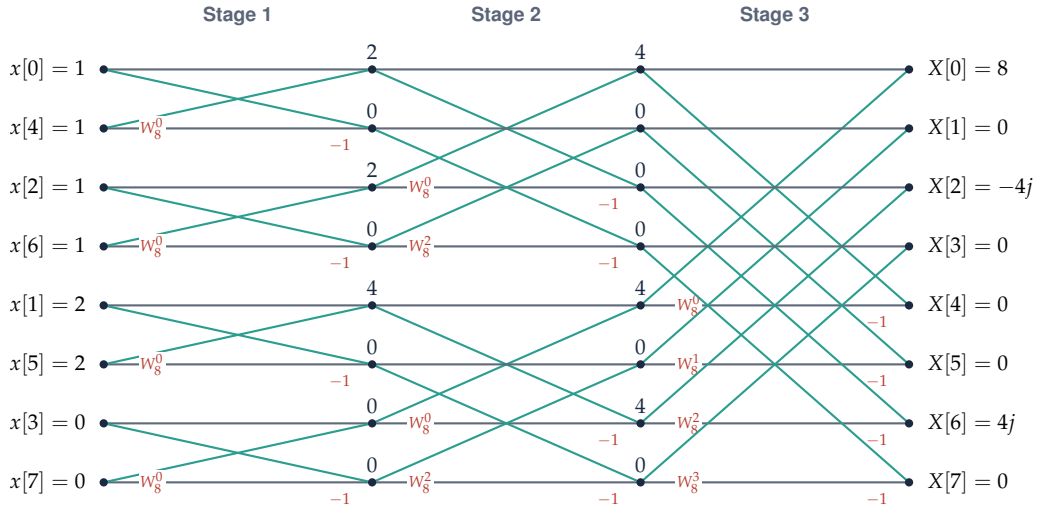
Step 1 — Twiddle factors and input order. $W_8 = e^{-j\pi/4}$: $W_8^0 = 1$, $W_8^1 = \frac{1-j}{\sqrt{2}}$, $W_8^2 = -j$, $W_8^3 = \frac{-1-j}{\sqrt{2}}$. Decimation in time feeds the inputs in *bit-reversed* order:

n (binary)	000	100	010	110	001	101	011	111
input	$x[0]$	$x[4]$	$x[2]$	$x[6]$	$x[1]$	$x[5]$	$x[3]$	$x[7]$
value	1	1	1	1	2	2	0	0

Step 2 — The three butterfly stages. Each butterfly computes $(a, b) \mapsto (a + Wb,\ a - Wb)$.

- **Stage 1** (2-point DFTs, $W = 1$): $(1,1) \to (2,0)$, $(1,1) \to (2,0)$, $(2,2) \to (4,0)$, $(0,0) \to (0,0)$.
- **Stage 2** (4-point, $W_4^0 = 1$, $W_4^1 = W_8^2 = -j$): top half $[2,0,2,0] \to [4,\ 0,\ 0,\ 0]$; bottom half $[4,0,0,0] \to [4,\ 0,\ 4,\ 0]$.
- **Stage 3** (8-point, $W_8^0..W_8^3$): $X[k] = G[k] + W_8^k H[k]$, $X[k+4] = G[k] - W_8^k H[k]$ with $G = [4,0,0,0]$, $H = [4,0,4,0]$:

$$\begin{aligned} X[0] &= G[0] + H[0] = 4 + 4 = 8, & X[4] &= 4 - 4 = 0, \\ X[1] &= 0 + W_8^1(0) = 0, & X[5] &= 0, \\ X[2] &= 0 + (-j)(4) = -4j, & X[6] &= 0 - (-j)(4) = 4j, \\ X[3] &= 0 + W_8^3(0) = 0, & X[7] &= 0. \end{aligned}$$



(Upper branch of every butterfly: plain addition; lower branch: multiplied by the twiddle W_8^r shown in red, then subtracted. Bold numbers are intermediate values.)

$$\boxed{X[k] = \{8,\ 0,\ -4j,\ 0,\ 0,\ 0,\ 4j,\ 0\}}$$

Step 3 — Check. The block is $\{1,2,1,0\}$ repeated, i.e. $x[n] = 1 + \sin\left(\frac{\pi n}{2}\right)$: a d.c. value 1 plus a sinusoid at $\frac{f_s}{4} = 2$ kHz. The DFT of 1 is $N\delta[k] = 8\delta[k]$; the DFT of $\sin(2\pi \cdot 2n/8)$ is $\frac{N}{2j}$ at $k = 2$ and $-\frac{N}{2j}$ at $k = 6$, i.e. $-4j$ and $+4j$. ✓ Also $X[0] = \sum x[n] = 8$ and $X[8-k] = X^*[k]$ (real input).

Step 4 — Which bins carry the vocal range? Bin k corresponds to frequency

$$f_k = k\,\frac{f_s}{N} = k \cdot 1\ \text{kHz} \quad (k = 0..4), \qquad f_k = (k-8)\ \text{kHz} \quad (k = 5..7,\ \text{negative frequencies}).$$

k	0	1	2	3	4	5	6	7
f_k (kHz)	0	1	2	3	± 4	-3	-2	-1
in 0.3–3 kHz?	no (d.c.)	**yes**	**yes**	**yes**	no	**yes**	**yes**	**yes**
$X[k]$	8	0	$-4j$	0	0	0	$4j$	0

With a resolution of $f_s/N = 1$ kHz the vocal range 0.3–3 kHz is represented by bins $k = 1, 2, 3$; because the signal is real, their negative-frequency partners $k = 7, 6, 5$ must be scaled by the *same* real gain, otherwise $X[8-k] = X^*[k]$ breaks and the inverse FFT returns a complex signal. Bin 0 (d.c.) and bin 4 (the Nyquist frequency, 4 kHz) are left alone. (The part 300 Hz–1 kHz falls between bins 0 and 1; with only 8 points the resolution cannot separate it more finely.)
For this particular test block, only $X[2] = -4j$ and $X[6] = 4j$ are non-zero in that range: the 2 kHz tone is the only vocal-range content, so boosting by G gives $x_{\text{out}}[n] = 1 + G\sin(\pi n/2)$.

Answer. $X = \{8, 0, -4j, 0, 0, 0, 4j, 0\}$. The equaliser must scale bins $k = 1, 2, 3$ and their mirror bins $k = 5, 6, 7$ (1, 2, 3 kHz) by the same gain; for this block only bins 2 and 6 (the 2 kHz component) are non-zero.

02

The Parking Sensor: Band-pass Sampling and Cross-correlation

QUESTION 2 • 35 MARKS

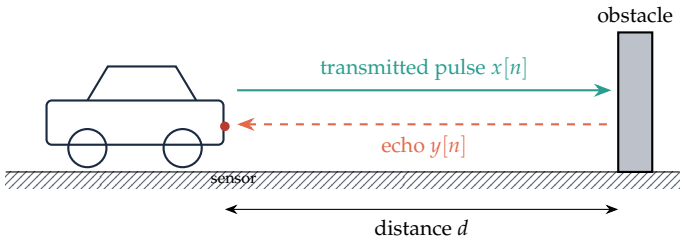
THE SETTING

A parking sensor sends out a short 40 kHz ultrasonic pulse, listens for the echo from the nearest obstacle, and works out the distance (Figure 1 of the paper). Because of the Doppler effect and other effects, the received signal occupies the band 38–42 kHz. The Nyquist rate for this signal is therefore 84 kHz, but the sensor samples at a far lower rate, $f_s = 18$ kHz. The engineer argues that, in the sampled signal, the copies of the positive and negative halves of the spectrum do not overlap.

Once the echo is captured, its delay is found by cross-correlating the received samples $y[n]$ with the stored samples $x[n]$ of the transmitted pulse,

$$c[m] = \sum_{n} x[n]\, y[n+m],$$

which has its peak at the delay. The software computes $c[m]$ through the DFT. The delay in samples gives the distance, remembering that the sound travels to the obstacle and back at 340 m/s.



Q2(a) Is the engineer right? (band-pass sampling) MARKS 12

Examine, with a sketch of the spectrum of the received signal sampled at $f_s = 18$ kHz, whether the engineer's claim is correct and where the 38–42 kHz band appears in the baseband $|f| < 9$ kHz.

SOLUTION

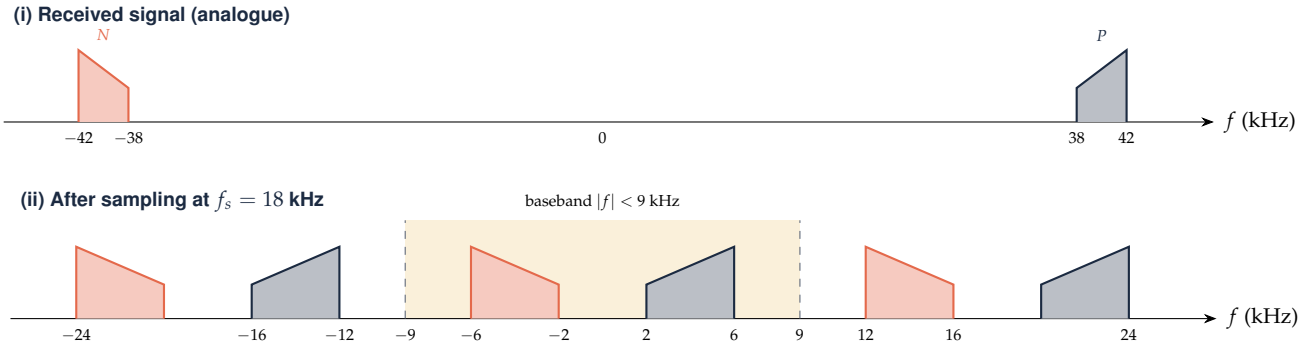
Step 1 — The idea. The Nyquist rate $2f_{\max} = 84$ kHz is needed for a signal that occupies *all* frequencies up to 42 kHz. This signal occupies only a 4 kHz-wide band, $B = 42 - 38 = 4$ kHz. Sampling still repeats the spectrum every f_s; what matters is only whether the shifted copies of

the band *collide*.

Step 2 — Where the copies go. The real signal has two bands: $P = [38, 42]$ kHz and its mirror $N = [-42, -38]$ kHz. After sampling, each is copied to every shift by $18k$ kHz:

shift	copy of $P = [38, 42]$	copy of $N = [-42, -38]$
-18 / $+18$	$[20, 24]$	$[-24, -20]$
-36 / $+36$	$[\mathbf{2}, \mathbf{6}]$	$[-\mathbf{6}, -\mathbf{2}]$
-54 / $+54$	$[-16, -12]$	$[12, 16]$

Inside the baseband $|f| < 9$ kHz we find exactly two copies: $[2, 6]$ kHz (from the positive band) and $[-6, -2]$ kHz (from the negative band). They do not touch: the gap between them is 4 kHz around $f = 0$, and 3 kHz to each edge ± 9.



Blue: copies of the positive band P; orange: copies of the negative band N. The sloped tops show the orientation (38 kHz edge low, 42 kHz edge high).

Step 3 — Formal check (band-pass sampling theorem). A band $[f_L, f_H]$ sampled at f_s is alias-free if for some integer n

$$\frac{2f_H}{n} \le f_s \le \frac{2f_L}{n-1}.$$

With $f_L = 38$, $f_H = 42$: for $n = 5$, $\frac{84}{5} = 16.8 \le 18 \le \frac{76}{4} = 19$ ✓. So $f_s = 18$ kHz lies inside an allowed window.

Step 4 — Where the band lands. 38 kHz maps to $38 - 2(18) = 2$ kHz and 42 kHz to 6 kHz: the band is translated, *not* inverted (an even multiple of $f_s/2$ was subtracted). A Doppler echo at 40 kHz $\pm\Delta$ appears at 4 kHz $\pm\Delta$.

> **KEY IDEA** Band-pass (under-)sampling: a narrow band far from d.c. only needs $f_s > 2B$ *and* a placement where the copies of $+$ and $-$ halves interleave without touching.

Answer. The engineer is right. At $f_s = 18$ kHz the band 38–42 kHz appears in the baseband at 2–6 kHz (positive half) and at -6 to -2 kHz (negative half); the copies do not overlap ($16.8 \le f_s \le 19$ kHz is an allowed band-pass window).

Q2(b) Cross-correlation through the DFT — MARKS 12

Demonstrate that the circular cross-correlation $c[m] = \sum_{n=0}^{N-1} x[n]\, y[((n+m))_N]$ of two real length-N sequences has the DFT $C[k] = X^*[k]\, Y[k]$, stating how much zero-padding, if any, is

needed for this route to give the ordinary (linear) cross-correlation.

SOLUTION

Step 1 — Take the DFT of $c[m]$. With $W = W_N = e^{-j2\pi/N}$,

$$C[k] = \sum_{m=0}^{N-1} c[m]W^{km} = \sum_{n=0}^{N-1} x[n] \sum_{m=0}^{N-1} y[((n+m))_N]\, W^{km}.$$

Step 2 — Change the inner index. Put $l = ((n+m))_N$. As m runs over $0..N-1$, so does l, and $m \equiv l - n \pmod N$. Since $W^N = 1$, $W^{km} = W^{k(l-n)} = W^{kl}W^{-kn}$:

$$C[k] = \sum_{n=0}^{N-1} x[n]W^{-kn} \sum_{l=0}^{N-1} y[l]W^{kl}.$$

Step 3 — Recognise the two factors. The second sum is $Y[k]$. For a *real* x, $\sum x[n]W^{-kn} = \left(\sum x[n]W^{kn}\right)^* = X^*[k]$. Hence

$$\boxed{C[k] = X^*[k]\, Y[k]} \qquad \Longrightarrow \qquad c = \text{IDFT}\{X^*\, Y\}.$$

Cost: two FFTs, one product and one inverse FFT, $O(N \log N)$, against $O(N^2)$ for the direct double sum.

Step 4 — Zero-padding. The linear cross-correlation $c_{\text{lin}}[m] = \sum_n x[n]y[n+m]$ of a length-N_x pulse and a length-N_y recording is non-zero for lags

$$-(N_x - 1) \le m \le N_y - 1 \qquad (N_x + N_y - 1 \text{ values}).$$

A length-M circular correlation equals the linear one *folded* with period M: $c_{\text{circ}}[m] = \sum_r c_{\text{lin}}[m + rM]$. The folded copies do not overlap only if $M \ge N_x + N_y - 1$. Therefore pad both sequences with zeros to

$$\boxed{M \ge N_x + N_y - 1 \;\; (= 2N - 1 \text{ for two length-}N \text{ sequences})}$$

Then $c_{\text{circ}}[m] = c_{\text{lin}}[m]$ for $0 \le m \le N_y - 1$, and the negative lags appear at the end: $c_{\text{circ}}[M - l] = c_{\text{lin}}[-l]$. In practice M is rounded up to a power of 2 for the radix-2 FFT.

Answer. $C[k] = \sum_n x[n]W^{-kn} \sum_l y[l]W^{kl} = X^*[k]Y[k]$. Zero-pad both sequences to at least $N_x + N_y - 1$ points ($2N - 1$ for equal lengths N) so the circular correlation equals the linear one.

Q2(c) Peak of $c[m]$ and the distance — MARKS 11

To save memory, the sensor only starts recording 30 samples after the pulse has been sent. For one measurement the stored pulse is $x = \{1, 2, 0, 0\}$ and the recording is $y = \{0, 1, 2, 0\}$. Use either method to find the peak of $c[m]$ and, from it, the distance to the obstacle.

SOLUTION

Method 1 — direct sum. $c[m] = \sum_n x[n]y[n+m]$, with only $x[0] = 1$, $x[1] = 2$ non-zero: $c[m] = y[m] + 2y[m+1]$.

lag m	-1	0	1	2	3
$y[m]+2y[m+1]$	$0+2(0)$	$0+2(1)$	$1+2(2)$	$2+2(0)$	0
$c[m]$	0	2	$\mathbf{5}$	2	0

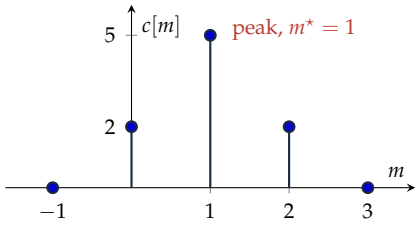
Method 2 — via the DFT. $N=4$ is enough: the pulse spans 2 samples and the recording 3, so $2+3-1=4$ (part (b)).

$$\begin{aligned} X[k] &= \{3,\ 1-2j,\ -1,\ 1+2j\},\\ Y[k] &= \{3,\ -2-j,\ 1,\ -2+j\},\\ C[k] = X^*[k]\,Y[k] &= \{9,\ (1+2j)(-2-j),\ (-1)(1),\ (1-2j)(-2+j)\} = \{9,\ -5j,\ -1,\ 5j\}. \end{aligned}$$

Inverse DFT, $c[m]=\frac{1}{4}\sum_k C[k]\,j^{km}$:

$$c[0]=\tfrac{9-5j-1+5j}{4}=2,\quad c[1]=\tfrac{9+5+1+5}{4}=5,\quad c[2]=\tfrac{9+5j-1-5j}{4}=2,\quad c[3]=\tfrac{9-5+1-5}{4}=0.$$

Both methods agree: $c=\{2,\mathbf{5},2,0\}$, peak at lag $m^\star=1$ (the recorded $\{1,2\}$ is the pulse shifted by one sample).



Step 3 — Total delay. The recording started 30 samples after transmission, so the echo delay is

$$D = 30 + m^\star = 31 \text{ samples}, \qquad \tau = \frac{D}{f_s} = \frac{31}{18\,000}\ \text{s} = 1.722\ \text{ms}.$$

Step 4 — Distance. The sound goes to the obstacle and back: $2d = v\tau$.

$$d = \frac{v\tau}{2} = \frac{340 \times 31}{2 \times 18\,000} = 0.293\ \text{m}.$$

Answer. $c[m]=\{2,5,2,0\}$ ($m=0..3$), peak $c[1]=5$; total delay 31 samples $=1.722$ ms at $f_s=18$ kHz, so the obstacle is $d\approx 0.293$ m ≈ 29 cm away.

03

The Bank Loan as a Discrete-time System

QUESTION 3 • 35 MARKS

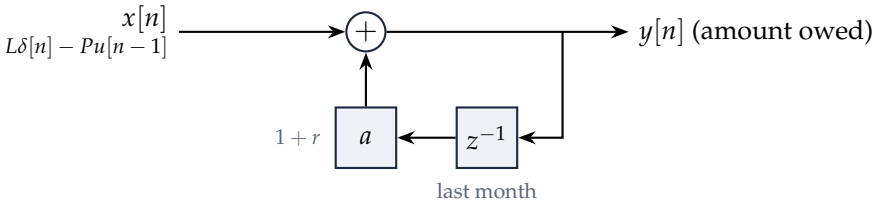
THE SETTING

A bank lends L taka, to be repaid in N equal monthly instalments (EMI) of P taka each, at a monthly interest rate r. At the end of every month the bank adds interest at rate r to the amount still owed, and the borrower then pays the instalment P. If $y[n]$ is the amount owed just after the n-th payment, then $y[0] = L$ and

$$y[n] = (1+r)\,y[n-1] - P, \qquad n \geq 1.$$

Treating the money as a signal: the input is $x[n] = L\delta[n] - Pu[n-1]$ (loan received once at $n = 0$, payment every month $n \geq 1$), and the amount owed obeys the causal first-order system

$$y[n] = a\,y[n-1] + x[n], \qquad a = 1 + r.$$



Q3(a) z-transform of the amount owed MARKS 12

The z-transform of the amount owed follows from the z-transform of the input $x[n]$ and that of the first-order difference equation relating $y[n]$ to $x[n]$. **Demonstrate** that it is

$$Y(z) = \frac{L}{1 - az^{-1}} - \frac{Pz^{-1}}{(1 - z^{-1})(1 - az^{-1})}, \qquad |z| > a.$$

SOLUTION

Step 1 — Transform of the input. Using $\delta[n] \longleftrightarrow 1$ (all z) and the delay property on $u[n] \longleftrightarrow \frac{1}{1-z^{-1}}$ ($|z| > 1$):

$$X(z) = L - P\,\frac{z^{-1}}{1 - z^{-1}}, \qquad |z| > 1.$$

Step 2 — Transform of the system. The system is causal and starts from rest before $n = 0$ (nothing owed before the loan), so the delay property gives $\mathcal{Z}\{y[n-1]\} = z^{-1}Y(z)$:

$$Y(z) = az^{-1}Y(z) + X(z) \implies H(z) = \frac{Y(z)}{X(z)} = \frac{1}{1-az^{-1}}, \qquad |z| > a.$$

(Check: at $n = 0$, $y[0] = a\,y[-1] + x[0] = 0 + L = L$, as required.)

Step 3 — Multiply.

$$Y(z) = H(z)X(z) = \frac{L}{1-az^{-1}} - \frac{Pz^{-1}}{(1-z^{-1})(1-az^{-1})}.$$

Step 4 — ROC. The poles are $z = a$ and $z = 1$; y is right-sided, so the ROC is outside the outermost pole. With $a = 1 + r > 1$:

$$|z| > \max(1, a) = a.$$

Answer. $Y(z) = \dfrac{L}{1-az^{-1}} - \dfrac{Pz^{-1}}{(1-z^{-1})(1-az^{-1})}$, ROC $|z| > a$ (since $a > 1$).

Q3(b) Inverse z-transform: the closed form — MARKS 13

Demonstrate, by inverting this $Y(z)$ with its ROC $|z| > a$, that the amount owed is

$$y[n] = La^n - \frac{P}{r}(a^n - 1), \qquad n \geq 0.$$

SOLUTION

Step 1 — First term. $\dfrac{L}{1-az^{-1}}$, ROC $|z| > a$ (right-sided) $\implies La^n u[n]$.

Step 2 — Partial fractions for the second term. Write $w = z^{-1}$:

$$\frac{w}{(1-w)(1-aw)} = \frac{A}{1-w} + \frac{B}{1-aw}.$$

Cover-up: at $w = 1$, $A = \dfrac{1}{1-a}$; at $w = \frac{1}{a}$, $B = \dfrac{1/a}{1-1/a} = \dfrac{1}{a-1}$. Since $a - 1 = r$,

$$\frac{z^{-1}}{(1-z^{-1})(1-az^{-1})} = \frac{1}{r}\left[\frac{1}{1-az^{-1}} - \frac{1}{1-z^{-1}}\right].$$

(Check at $w = 0$: left side 0, right side $\frac{1}{r}(1-1) = 0$ ✓.)

Step 3 — Invert with the ROC. Both terms are right-sided ($|z| > a > 1$): $\frac{1}{1-az^{-1}} \to a^n u[n]$ and $\frac{1}{1-z^{-1}} \to u[n]$. Hence

$$\mathcal{Z}^{-1}\left\{\frac{z^{-1}}{(1-z^{-1})(1-az^{-1})}\right\} = \frac{a^n - 1}{r}\,u[n].$$

Step 4 — Combine.

$$\boxed{y[n] = La^n - \frac{P}{r}(a^n - 1), \qquad n \geq 0}$$

Check. $n=0$: $y[0]=L-0=L$ ✓. $n=1$: $La-\frac{P}{r}(a-1)=L(1+r)-P$ ✓ (one month of interest, one payment). Interpretation: the loan grows like La^n while the payments accumulate like a savings account, $P\frac{a^n-1}{r}=P(1+a+\cdots+a^{n-1})$.

Answer. $y[n]=La^n-\dfrac{P}{r}(a^n-1)$, $n\geq 0$ (from $Y=\frac{L}{1-az^{-1}}-\frac{P}{r}\left[\frac{1}{1-az^{-1}}-\frac{1}{1-z^{-1}}\right]$).

Q3(c) The EMI formula

MARKS 10

The loan is fully repaid when $y[N]=0$. **Examine** this condition on the closed form of part (b) to obtain the EMI P in terms of L, r and N, and its value for $L=5{,}00{,}000$ taka, $r=1\%$ per month and $N=24$ months.

SOLUTION

Step 1 — Set $y[N]=0$.

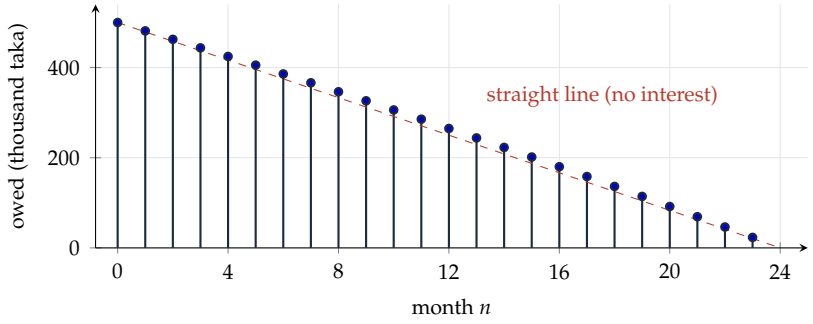
$$La^N-\frac{P}{r}(a^N-1)=0 \quad\Longrightarrow\quad \boxed{P=\frac{L\,r\,(1+r)^N}{(1+r)^N-1}}$$

This is the standard EMI formula of banking.

Step 2 — Numbers. $L=500\,000$, $r=0.01$, $N=24$: $(1.01)^{24}=1.269735$.

$$P=\frac{500\,000\times 0.01\times 1.269735}{1.269735-1}=\frac{6348.67}{0.269735}=23\,536.74 \text{ taka.}$$

Step 3 — Sanity checks. Total repaid $=24P=564\,882$ taka, i.e. 64 882 taka of interest. Without interest the instalment would be $500\,000/24=20\,833$; the EMI must be larger, and it is. Iterating $y[n]=1.01\,y[n-1]-P$ from $y[0]=500\,000$ with this P gives $y[24]=0$ (to rounding).



The balance falls slowly at first (most of each early instalment pays interest) and faster near the end — the curve lies above the straight line.

Answer. $P=\dfrac{Lr(1+r)^N}{(1+r)^N-1}$; for $L=5{,}00{,}000$, $r=1\%$, $N=24$: $P\approx 23{,}537$ taka per month (total repaid $\approx 5{,}64{,}882$ taka).

04

The Car Suspension: Laplace Transform and Damping

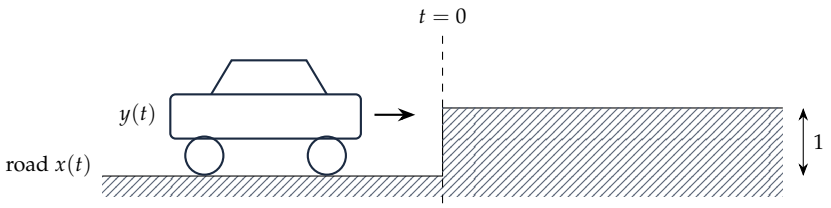
QUESTION 4 • 35 MARKS

THE SETTING

In a simple model of a car's suspension, the height of the road surface $x(t)$ drives the height of the car body $y(t)$ through

$$y'' + b\,y' + 10\,y = 10\,x,$$

with units chosen so that the spring constant is 10. The constant b is the setting of the shock absorber (damper): a larger b means a harder damper, a smaller b a softer one. A soft damper gives a bouncy ride and a hard one a slow, stiff ride. The test is the car driving onto a step in the road, $x(t) = u(t)$ (Figure 2 of the paper).



KEY IDEA Laplace-transform the equation with zero initial conditions: $H(s) = \dfrac{10}{s^2 + bs + 10}$. For the step, $Y(s) = \dfrac{10}{s(s^2 + bs + 10)}$. The poles of H decide everything: real ⇒ no oscillation, complex ⇒ bounce. Since $H(0) = 1$, every stable setting settles at $y(\infty) = 1$, the height of the step.

Q4(a) Hard damper, $b = 7$ MARKS 12

Solve for the response $y(t)$ of the car body with the damper set hard, $b = 7$, and find out whether the body ever rises above the height of the step (that is, above 1).

SOLUTION

Step 1 — Transform. $s^2 + 7s + 10 = (s+2)(s+5)$, so

$$Y(s) = \frac{10}{s(s+2)(s+5)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+5}.$$

Step 2 — Cover-up.

$$A = \frac{10}{(2)(5)} = 1, \qquad B = \frac{10}{(-2)(3)} = -\frac{5}{3}, \qquad C = \frac{10}{(-5)(-3)} = \frac{2}{3}.$$

(Check: $A + B + C = 0 = y(0)$ ✓.)

Step 3 — Invert.

$$\boxed{y(t) = \left(1 - \tfrac{5}{3}e^{-2t} + \tfrac{2}{3}e^{-5t}\right)u(t)}$$

Step 4 — Does it overshoot? Differentiate:

$$y'(t) = \tfrac{10}{3}e^{-2t} - \tfrac{10}{3}e^{-5t} = \tfrac{10}{3}e^{-2t}\left(1 - e^{-3t}\right) > 0 \quad \text{for all } t > 0.$$

y is strictly increasing and tends to 1, so it approaches 1 from below and *never exceeds it*. (Over-damped: two real poles, -2 and -5.)

Answer. $y(t) = 1 - \frac{5}{3}e^{-2t} + \frac{2}{3}e^{-5t}$ $(t \geq 0)$. Since $y'(t) > 0$ for all $t > 0$ and $y \to 1$, the body never rises above the step.

Q4(b) Soft damper, $b = 2$ MARKS 13

Solve for the response $y(t)$ of the car body with the damper set soft, $b = 2$, and find out the frequency at which the body bounces.

SOLUTION

Step 1 — Poles. $s^2 + 2s + 10 = (s+1)^2 + 9$: complex poles $s = -1 \pm j3$.

Step 2 — Partial fractions.

$$Y(s) = \frac{10}{s\left[(s+1)^2 + 9\right]} = \frac{A}{s} + \frac{Bs + C}{(s+1)^2 + 9}, \qquad A = \frac{10}{10} = 1.$$

Then $10 = (s^2 + 2s + 10) + (Bs + C)s \Rightarrow s^2$: $1 + B = 0 \Rightarrow B = -1$; s^1: $2 + C = 0 \Rightarrow C = -2$. So

$$Y(s) = \frac{1}{s} - \frac{s+2}{(s+1)^2 + 9} = \frac{1}{s} - \frac{s+1}{(s+1)^2 + 3^2} - \frac{1}{3}\cdot\frac{3}{(s+1)^2 + 3^2}.$$

Step 3 — Invert with $e^{-at}\cos\omega t \longleftrightarrow \frac{s+a}{(s+a)^2+\omega^2}$, $e^{-at}\sin\omega t \longleftrightarrow \frac{\omega}{(s+a)^2+\omega^2}$:

$$\boxed{y(t) = \left[1 - e^{-t}\left(\cos 3t + \tfrac{1}{3}\sin 3t\right)\right]u(t)}$$

(Check: $y(0) = 1 - 1 = 0$ ✓. Differentiating, $y'(t) = e^{-t}\left(\cos 3t + \frac{1}{3}\sin 3t\right) - e^{-t}\left(\cos 3t - 3\sin 3t\right) = \frac{10}{3}e^{-t}\sin 3t$, so $y'(0) = 0$ ✓— the body starts at rest.)

Step 4 — Bounce frequency. The oscillation is $\cos 3t$ inside a decaying envelope e^{-t}: the *damped* natural frequency is the imaginary part of the poles,

$$\omega_d = 3 \text{ rad/s}, \qquad f_d = \frac{3}{2\pi} = 0.477 \text{ Hz}, \qquad T_d = \frac{2\pi}{3} = 2.09 \text{ (time units)}.$$

The first peak is where $y' = 0$, i.e. $\sin 3t = 0$, $t = \pi/3$: $y(\pi/3) = 1 + e^{-\pi/3} = 1.351$, a 35% overshoot above the step.

Answer. $y(t) = 1 - e^{-t}(\cos 3t + \frac{1}{3}\sin 3t)$; the body bounces at $\omega_d = 3$ rad/s ($f \approx 0.48$ Hz), first peak 1.35 at $t = \pi/3$.

Q4(c) Critical damping MARKS 10

The car body oscillates when the poles are complex and does not when they are real. **Examine** the poles for the smallest value of b at which the oscillation disappears, so that the car body reaches the height of the step as fast as possible without going above it.

SOLUTION

Step 1 — Poles as a function of b.

$$s^2 + bs + 10 = 0 \quad \Longrightarrow \quad s = \frac{-b \pm \sqrt{b^2 - 40}}{2}.$$

Complex (oscillation) when $b^2 < 40$; real (no oscillation) when $b^2 \geq 40$.

Step 2 — Boundary. The smallest b with real poles is

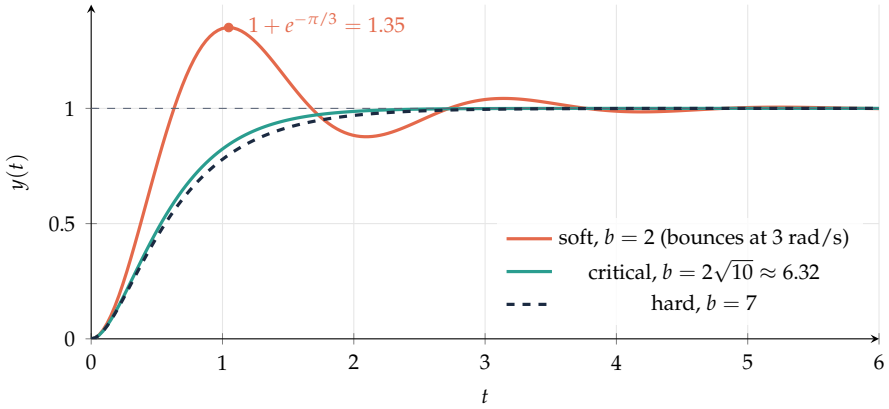
$$b^2 = 40 \;\Rightarrow\; \boxed{b = 2\sqrt{10} \approx 6.32}, \qquad s_{1,2} = -\frac{b}{2} = -\sqrt{10} \approx -3.16 \text{ (a double pole).}$$

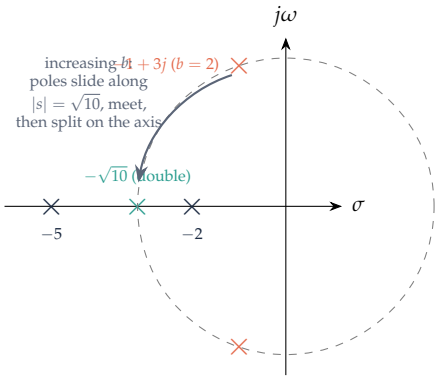
This is *critical damping* (damping ratio $\zeta = \frac{b}{2\sqrt{10}} = 1$).

Step 3 — Response. $Y(s) = \dfrac{10}{s(s+\sqrt{10})^2} = \dfrac{1}{s} - \dfrac{1}{s+\sqrt{10}} - \dfrac{\sqrt{10}}{(s+\sqrt{10})^2}$, so

$$y(t) = 1 - \left(1 + \sqrt{10}\,t\right)e^{-\sqrt{10}\,t}, \qquad y'(t) = 10\,t\,e^{-\sqrt{10}t} > 0.$$

Monotonic, never above 1. For $b < 2\sqrt{10}$ the poles become complex and the body overshoots; for $b > 2\sqrt{10}$ one pole moves towards the origin (e.g. -2 for $b = 7$), making the response slower. So $b = 2\sqrt{10}$ gives the fastest rise without overshoot.





Answer. Oscillation disappears when $b^2 - 40 \geq 0$; the smallest such value is $b = 2\sqrt{10} \approx 6.32$ (critical damping), with a double real pole at $s = -\sqrt{10} \approx -3.16$ and $y(t) = 1 - (1 + \sqrt{10}t)e^{-\sqrt{10}t}$ — the fastest response with no overshoot.

Section B

Questions 5–8

Signals • Convolution • Fourier series • Fourier transform

05

Signals: Even–Odd Parts, Causality and the Fourier Series

QUESTION 5 • MANDATORY • 35 MARKS

Q5(a) Sketch and even–odd decomposition

MARKS 4 + 8

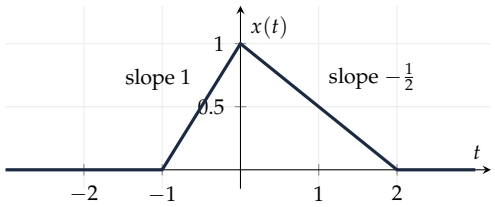
Consider the continuous-time signal

$$x(t) = \begin{cases} t+1, & -1 \le t \le 0, \\ 1 - \dfrac{t}{2}, & 0 < t \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

(i) Sketch the signal $x(t)$. (ii) Decompose $x(t)$ into its even part $x_e(t)$ and odd part $x_o(t)$, analytically or graphically, and sketch $x_e(t)$ and $x_o(t)$.

SOLUTION

(i) The signal. A triangle: it rises with slope $+1$ from 0 at $t = -1$ to the peak 1 at $t = 0$, then falls with slope $-\frac{1}{2}$ to 0 at $t = 2$. (It is continuous: both pieces give $x(0) = 1$.)



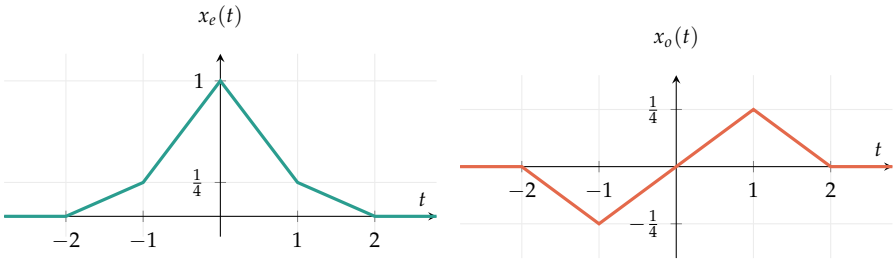
(ii) Formulas. $x_e(t) = \dfrac{x(t) + x(-t)}{2}$, $x_o(t) = \dfrac{x(t) - x(-t)}{2}$. It is enough to work for $t \ge 0$ and then use symmetry (x_e even, x_o odd). For $t \ge 0$, $x(-t)$ uses the first piece: $x(-t) = 1 - t$ for $0 \le t \le 1$ and 0 for $t > 1$.

interval	$x(t)$	$x(-t)$	$x_e(t)$	$x_o(t)$
$0 \le t \le 1$	$1 - \frac{t}{2}$	$1 - t$	$1 - \frac{3t}{4}$	$\frac{t}{4}$
$1 \le t \le 2$	$1 - \frac{t}{2}$	0	$\frac{1}{2} - \frac{t}{4}$	$\frac{1}{2} - \frac{t}{4}$
$t \ge 2$	0	0	0	0

For $t<0$: $x_e(t)=x_e(-t)$ and $x_o(t)=-x_o(-t)$. In full:

$$x_e(t)=\begin{cases}1-\frac{3}{4}|t|, & |t|\le 1\\ \frac{1}{2}-\frac{|t|}{4}, & 1\le|t|\le 2\\ 0, & |t|\ge 2\end{cases}\qquad x_o(t)=\begin{cases}\frac{t}{4}, & |t|\le 1\\ \operatorname{sgn}(t)\left(\frac{1}{2}-\frac{|t|}{4}\right), & 1\le|t|\le 2\\ 0, & |t|\ge 2\end{cases}$$

Key values: $x_e(0)=1$, $x_e(\pm1)=\frac{1}{4}$, $x_e(\pm2)=0$; $x_o(0)=0$, $x_o(\pm1)=\pm\frac{1}{4}$, $x_o(\pm2)=0$.



Check. x_e+x_o on $0\le t\le 1$: $1-\frac{3t}{4}+\frac{t}{4}=1-\frac{t}{2}$ ✓; on $-1\le t\le 0$: $1-\frac{3}{4}(-t)+\frac{t}{4}=1+t$ ✓; on $1\le t\le 2$: $2(\frac{1}{2}-\frac{t}{4})=1-\frac{t}{2}$ ✓; on $-2\le t\le -1$: $(\frac{1}{2}+\frac{t}{4})-(\frac{1}{2}+\frac{t}{4})=0$ ✓.

Answer. $x_e(t)$: triangle-like even signal with $x_e(0)=1$, $x_e(\pm1)=\frac{1}{4}$, zero for $|t|\ge 2$; $x_o(t)$: odd signal rising linearly to $\frac{1}{4}$ at $t=1$, back to 0 at $t=2$ (mirrored negative for $t<0$).

Q5(b) Minimum shift for causality — MARKS 6

Let $x(t)=e^{-3t}u(t+3)$ and $y(t)=x(t-a)$. Identify the minimum value of a for which $y(t)$ is a causal signal.

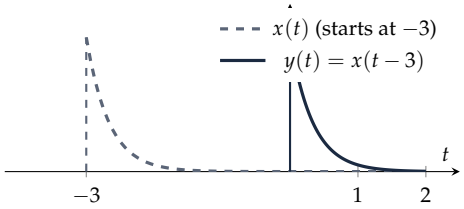
SOLUTION

Step 1 — Support of x. $u(t+3)=1$ for $t\ge -3$, so $x(t)\neq 0$ for $t\ge -3$: x starts at $t=-3$ (with value e^9) and is *not* causal.

Step 2 — Support of y. $y(t)=x(t-a)=e^{-3(t-a)}u(t-a+3)$ is non-zero for $t-a+3\ge 0$, i.e. $t\ge a-3$.

Step 3 — Causal condition. A causal signal is zero for $t<0$, so we need $a-3\ge 0$:

$$a\ge 3 \implies a_{\min}=3,\qquad y(t)=e^{-3(t-3)}u(t)=e^9e^{-3t}u(t).$$



(Heights normalised: both curves start at e^9; the shift moves the starting edge from $t=-3$ to $t=0$.)

Answer. $a_{\min}=3$ (a delay of 3 moves the starting point $t=-3$ to $t=0$).

Q5(c) Can a causal impulse response be periodic? MARKS 5

Consider an LTI system with impulse response $h(t)$. If the system is causal, can $h(t)$ be periodic? Explain your answer.

SOLUTION

Causal means $h(t) = 0$ for all $t < 0$. Suppose h is also periodic with some period $T > 0$: $h(t) = h(t - nT)$ for every integer n.
Take any time t. Choose n so large that $t - nT < 0$. Then

$$h(t) = h(t - nT) = 0 \quad \text{(causality)}.$$

Since t was arbitrary, $h(t) = 0$ for all t. So the only causal periodic impulse response is the trivial one, $h \equiv 0$ (a system that outputs nothing).
Intuition: periodicity forces the "past" $t < 0$ to repeat the "future" $t > 0$, while causality forces the past to be zero; both can hold only if everything is zero. (Also, a non-zero periodic h is never absolutely integrable, so it could not be stable either.)

Answer. No. A causal and periodic $h(t)$ must be identically zero: for any t, $h(t) = h(t - nT)$ with $t - nT < 0$, which is 0 by causality.

Q5(d) Complex exponential Fourier series MARKS 12

Consider the continuous-time periodic signal

$$x(t) = 4 + \sin\left(\omega_0 t - \frac{\pi}{4}\right) + 2\cos(3\omega_0 t).$$

Express $x(t)$ in complex exponential Fourier series form, and relate each non-zero coefficient to the corresponding frequency component of $x(t)$.

SOLUTION

Step 1 — Use Euler's formulas $\sin\theta = \dfrac{e^{j\theta} - e^{-j\theta}}{2j}$ and $\cos\theta = \dfrac{e^{j\theta} + e^{-j\theta}}{2}$. No integration is needed.

$$\sin\left(\omega_0 t - \frac{\pi}{4}\right) = \frac{e^{-j\pi/4}}{2j} e^{j\omega_0 t} - \frac{e^{j\pi/4}}{2j} e^{-j\omega_0 t}, \qquad 2\cos 3\omega_0 t = e^{j3\omega_0 t} + e^{-j3\omega_0 t}.$$

Step 2 — Simplify the coefficients. With $\frac{1}{j} = e^{-j\pi/2}$:

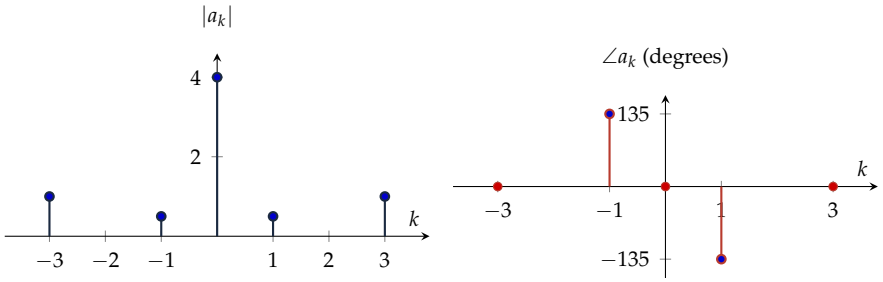
$$a_1 = \frac{e^{-j\pi/4}}{2j} = \frac{1}{2} e^{-j3\pi/4} = -\frac{\sqrt{2}}{4}(1 + j), \qquad a_{-1} = -\frac{e^{j\pi/4}}{2j} = \frac{1}{2} e^{j3\pi/4} = -\frac{\sqrt{2}}{4}(1 - j) = a_1^*.$$

Step 3 — The series.

$$\boxed{x(t) = 4 + \tfrac{1}{2} e^{-j3\pi/4} e^{j\omega_0 t} + \tfrac{1}{2} e^{j3\pi/4} e^{-j\omega_0 t} + e^{j3\omega_0 t} + e^{-j3\omega_0 t}}$$

k	a_k	$\lvert a_k\rvert$	$\angle a_k$	frequency component
0	4	4	0	d.c. (average) value 4
±1	$\frac{1}{2}e^{\mp j3\pi/4}$	$\frac{1}{2}$	$\mp135^\circ$	fundamental ω_0: $2\lvert a_1\rvert\cos(\omega_0 t+\angle a_1)=\cos(\omega_0 t-\frac{3\pi}{4})=\sin(\omega_0 t-\frac{\pi}{4})$
±3	1	1	0	third harmonic $3\omega_0$: $2\lvert a_3\rvert\cos 3\omega_0 t=2\cos 3\omega_0 t$
others	0	0	–	absent (no 2nd, 4th, … harmonic)

Step 4 — How the coefficients relate to the components. Each real sinusoid $A\cos(k\omega_0 t+\theta)$ splits into a conjugate pair $a_{\pm k}=\frac{A}{2}e^{\pm j\theta}$: the *magnitude* $\lvert a_k\rvert$ is half the amplitude, the *angle* is the phase, and $a_{-k}=a_k^*$ because x is real. The constant term is a_0.



Check (Parseval). $\sum\lvert a_k\rvert^2=16+2(\frac{1}{4})+2(1)=18.5$, and the average power directly: $4^2+\frac{1^2}{2}+\frac{2^2}{2}=18.5$ ✓.

Answer. $a_0=4$, $a_{\pm1}=\frac{1}{2}e^{\mp j3\pi/4}$, $a_{\pm3}=1$, all others 0: a_0 is the d.c. level, $a_{\pm1}$ the fundamental $\sin(\omega_0 t-\pi/4)$ (amplitude $2\lvert a_1\rvert=1$, phase -135° as a cosine) and $a_{\pm3}$ the third harmonic $2\cos 3\omega_0 t$.

06

Convolution, Causality and Stability

QUESTION 6 • 35 MARKS

Q6(a) Discrete convolution with a step MARKS 17

Let $0 < \alpha < 1$. Consider $x[n] = \alpha^n u[n]$ and $h[n] = u[n]$. Compute the convolution $y[n] = x[n] * h[n]$ and show that

$$y[n] = \frac{1 - \alpha^{n+1}}{1 - \alpha}\, u[n].$$

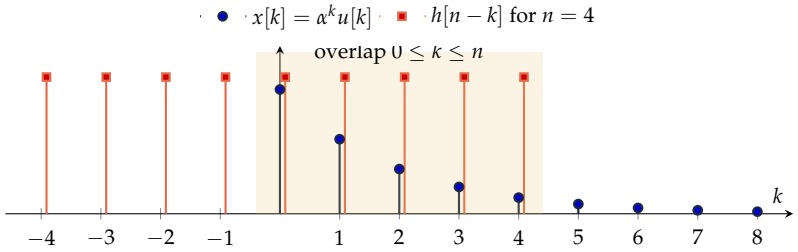
Also sketch $y[n]$.

SOLUTION

Step 1 — Definition.

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k] = \sum_{k=-\infty}^{\infty} \alpha^k u[k]\; u[n-k].$$

Step 2 — Find where the terms are non-zero. $u[k] = 1$ needs $k \geq 0$; $u[n-k] = 1$ needs $k \leq n$. (Graphically: $h[n-k]$ is the step flipped about $k = 0$ and slid to $k = n$; it covers $k \leq n$.)



- $n < 0$: no overlap (the step ends at $k = n < 0$ before x starts), so $y[n] = 0$.
- $n \geq 0$: the overlap is $0 \leq k \leq n$.

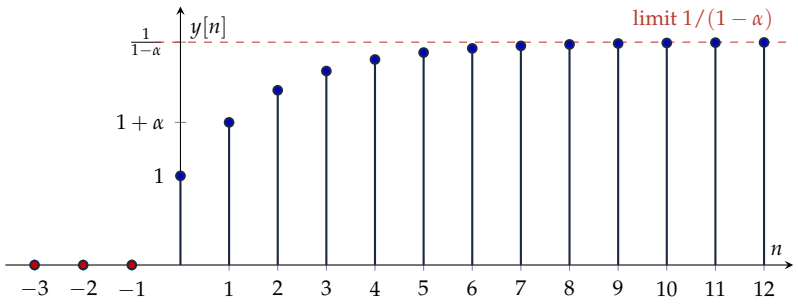
Step 3 — Sum the geometric series (first term 1, ratio $\alpha \neq 1$, $n+1$ terms):

$$y[n] = \sum_{k=0}^{n} \alpha^k = \frac{1 - \alpha^{n+1}}{1 - \alpha}, \qquad n \geq 0.$$

Combining both cases:

$$\boxed{y[n] = \frac{1 - \alpha^{n+1}}{1 - \alpha}\, u[n]}$$

Step 4 — Behaviour. $y[0] = 1, y[1] = 1+\alpha, y[2] = 1+\alpha+\alpha^2, \ldots$: convolving with $u[n]$ is a running sum (accumulator). Since $0 < \alpha < 1$, $\alpha^{n+1} \to 0$ and $y[n]$ rises monotonically to the limit $\dfrac{1}{1-\alpha}$.



NOTE The sketch is drawn for $\alpha = 0.6$, so $y[0] = 1$, $y[1] = 1.6$, $y[2] = 1.96$, …, $y[\infty] = 2.5$. The shape is the same for every $0 < \alpha < 1$: zero for $n < 0$, starts at 1, increases monotonically, saturates at $\frac{1}{1-\alpha}$.

Answer. $y[n] = \sum_{k=0}^{n} \alpha^k = \dfrac{1-\alpha^{n+1}}{1-\alpha} u[n]$; it starts at 1 and rises monotonically to $\dfrac{1}{1-\alpha}$.

Q6(b) Causality and stability of three systems **MARKS 6 + 6 + 6**

For each continuous-time LTI system determine whether it is causal and whether it is stable. Justify your answers.

$$\text{(i) } h_1(t) = e^{-3t}u(2-t) \qquad \text{(ii) } h_2(t) = e^{-5|t|} \qquad \text{(iii) } h_3(t) = te^{-2t}u(t)$$

SOLUTION

KEY IDEA For an LTI system: **causal** $\iff$ $h(t) = 0$ for all $t < 0$; **BIBO stable** $\iff$ $\int_{-\infty}^{\infty} |h(t)|\,dt < \infty$.

(i) $h_1(t) = e^{-3t}u(2-t)$. $u(2-t) = 1$ for $t \le 2$, so h_1 lives on $(-\infty, 2]$.

- Causal? $h_1(-1) = e^{3} \neq 0$: non-zero for negative time $\Rightarrow$ **not causal**.
- Stable? $\displaystyle\int_{-\infty}^{2} e^{-3t}\,dt = \Big[-\tfrac{1}{3}e^{-3t}\Big]_{-\infty}^{2} = -\tfrac{1}{3}e^{-6} + \tfrac{1}{3}\lim_{t\to-\infty} e^{-3t} = \infty$ (the exponential *grows* as $t \to -\infty$) $\Rightarrow$ **not stable**.

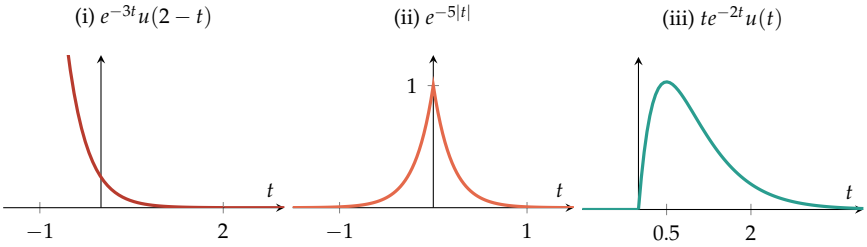
(ii) $h_2(t) = e^{-5|t|}$. A two-sided decaying exponential.

- Causal? $h_2(t) = e^{5t} \neq 0$ for $t < 0 \Rightarrow$ **not causal**.
- Stable? $\displaystyle\int_{-\infty}^{\infty} e^{-5|t|}\,dt = 2\int_{0}^{\infty} e^{-5t}\,dt = \frac{2}{5} < \infty \Rightarrow$ **stable**.

(iii) $h_3(t) = te^{-2t}u(t)$.

- Causal? $u(t)$ makes $h_3 = 0$ for $t < 0 \Rightarrow$ **causal**.

- Stable? $h_3 \geq 0$, and $\displaystyle\int_0^\infty te^{-2t}\,dt = \frac{1!}{2^2} = \frac{1}{4} < \infty$ (by parts, or $\int_0^\infty t^n e^{-at} = \frac{n!}{a^{n+1}}$) $\Rightarrow$ **stable**. The factor t grows only linearly; the exponential decay wins.



system	causal	stable	reason
(i) $e^{-3t}u(2-t)$	no	no	non-zero for $t<0$; grows as $t\to-\infty$
(ii) $e^{-5\lvert t\rvert}$	no	yes	non-zero for $t<0$; $\int\lvert h\rvert = \frac{2}{5}$
(iii) $te^{-2t}u(t)$	yes	yes	zero for $t<0$; $\int\lvert h\rvert = \frac{1}{4}$

Answer. (i) neither causal nor stable; (ii) stable but not causal ($\int|h| = \frac{2}{5}$); (iii) causal and stable ($\int|h| = \frac{1}{4}$).

07

Fourier Series: Properties, Impulse Trains and Derivatives

QUESTION 7 • 35 MARKS

Q7(a) Coefficients of $x(6-t)+y(t+2)$ MARKS 10

Suppose the periodic signals $x(t)$ and $y(t)$ have the same fundamental frequency ω_0, with complex exponential Fourier series coefficients $x(t) \leftrightarrow a_k$, $y(t) \leftrightarrow b_k$. Find the complex exponential Fourier series coefficients of $x(6-t)+y(t+2)$ in terms of a_k and b_k.

SOLUTION

Properties used (all with the same ω_0):

$$\text{time shift: } x(t-t_0) \leftrightarrow a_k e^{-jk\omega_0 t_0}, \qquad \text{time reversal: } x(-t) \leftrightarrow a_{-k}, \qquad \text{linearity.}$$

Step 1 — $x(6-t)$. Write $x(6-t) = x\big(-(t-6)\big)$: first reverse, then delay by 6.

$$x(-t) \leftrightarrow a_{-k} \quad \Longrightarrow \quad x\big(-(t-6)\big) \leftrightarrow a_{-k}\, e^{-jk\omega_0 \cdot 6}.$$

Direct check: $\frac{1}{T}\int x(6-t)e^{-jk\omega_0 t}\,dt$ with $\tau = 6-t$ gives $e^{-jk\omega_0 6}\frac{1}{T}\int x(\tau)e^{jk\omega_0\tau}\,d\tau = e^{-j6k\omega_0}a_{-k}$ ✓.

Step 2 — $y(t+2)$. An advance by 2 ($t_0 = -2$): $y(t+2) \leftrightarrow b_k\, e^{jk\omega_0 \cdot 2}$.

Step 3 — Add (linearity).

$$\boxed{c_k = a_{-k}\, e^{-j6k\omega_0} + b_k\, e^{j2k\omega_0}}$$

Answer. $x(6-t)+y(t+2) \leftrightarrow c_k = a_{-k}e^{-j6k\omega_0} + b_k e^{j2k\omega_0}$.

Q7(b) Fourier series of the impulse train MARKS 10

Consider the periodic impulse train $p(t) = \sum_{m=-\infty}^{\infty}\delta(t-mT)$. Using the sifting property of the impulse, show that its complex exponential Fourier series coefficients are $c_k = \frac{1}{T}$, $k = 0, \pm1, \pm2, \ldots$ Finally, sketch $p(t)$ against t and its spectrum c_k against $\omega = k\omega_0$ (where $\omega_0 = 2\pi/T$), and thereby show that an impulse train in the time domain remains an impulse train in the frequency domain.

SOLUTION

Step 1 — Choose a convenient period. Integrate over $-\frac{T}{2} < t < \frac{T}{2}$. Only the impulse at $m = 0$

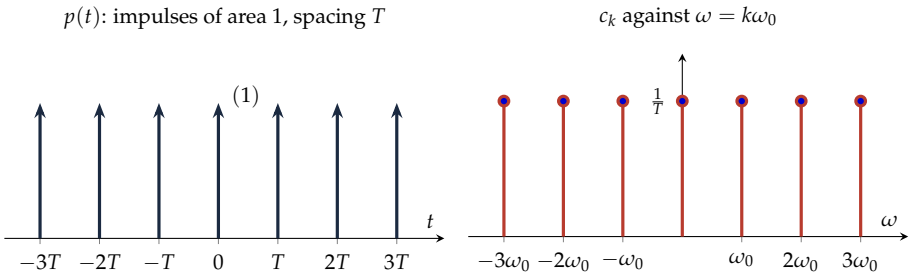
lies inside this window, so $p(t) = \delta(t)$ there.

Step 2 — Sifting. $\int f(t)\delta(t)\,dt = f(0)$:

$$c_k = \frac{1}{T}\int_{-T/2}^{T/2} \delta(t)\, e^{-jk\omega_0 t}\,dt = \frac{1}{T}\, e^{-jk\omega_0 \cdot 0} = \frac{1}{T} \qquad \text{for every } k.$$

So $p(t) = \dfrac{1}{T}\displaystyle\sum_{k=-\infty}^{\infty} e^{jk\omega_0 t}$: every harmonic is present with the same weight.

Step 3 — Sketches.



Step 4 — Impulse train → impulse train. The coefficients form a uniform comb: equal heights $\frac{1}{T}$ at equally spaced frequencies $k\omega_0$. As a Fourier transform each $e^{jk\omega_0 t} \longleftrightarrow 2\pi\delta(\omega - k\omega_0)$, so

$$P(j\omega) = \frac{2\pi}{T}\sum_{k=-\infty}^{\infty} \delta\Big(\omega - k\frac{2\pi}{T}\Big)\,:$$

an impulse train in time (spacing T) is an impulse train in frequency (spacing $2\pi/T$). Closer impulses in time ⇒ wider spacing in frequency.

Answer. $c_k = \frac{1}{T}\int_{-T/2}^{T/2}\delta(t)e^{-jk\omega_0 t}\,dt = \frac{1}{T}$ for all k; the spectrum is a comb of equal lines at $k\omega_0$, i.e. $P(j\omega) = \frac{2\pi}{T}\sum_k \delta(\omega - k\omega_0)$.

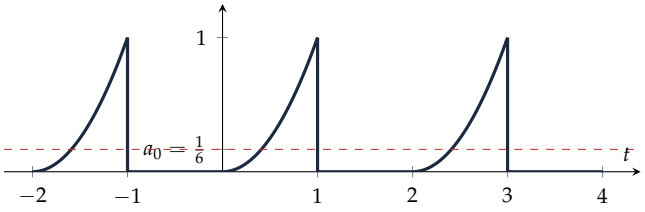
Q7(c) The d.c. coefficient — MARKS 8

Consider the periodic signal $x(t)$ with period $T = 2$, defined over one period as $x(t) = t^2$ for $0 \le t < 1$ and $x(t) = 0$ for $1 \le t < 2$. Let a_k be its Fourier series coefficients. Find a_0.

SOLUTION

a_0 is the average value over one period:

$$a_0 = \frac{1}{T}\int_0^T x(t)\,dt = \frac{1}{2}\Big[\int_0^1 t^2\,dt + \int_1^2 0\,dt\Big] = \frac{1}{2}\cdot\frac{1}{3} = \frac{1}{6}.$$



Answer. $a_0 = \dfrac{1}{6}$.

Q7(d) Sketch of $\frac{dx}{dt}$ MARKS 7

Consider another periodic signal $x(t)$ with period $T = 3$, defined over one period as $x(t) = \frac{t}{2}$ for $0 \le t < 2$ and $x(t) = 2 - t$ for $2 \le t < 3$. Sketch $\frac{dx(t)}{dt}$, labelling the important values on both axes.

SOLUTION

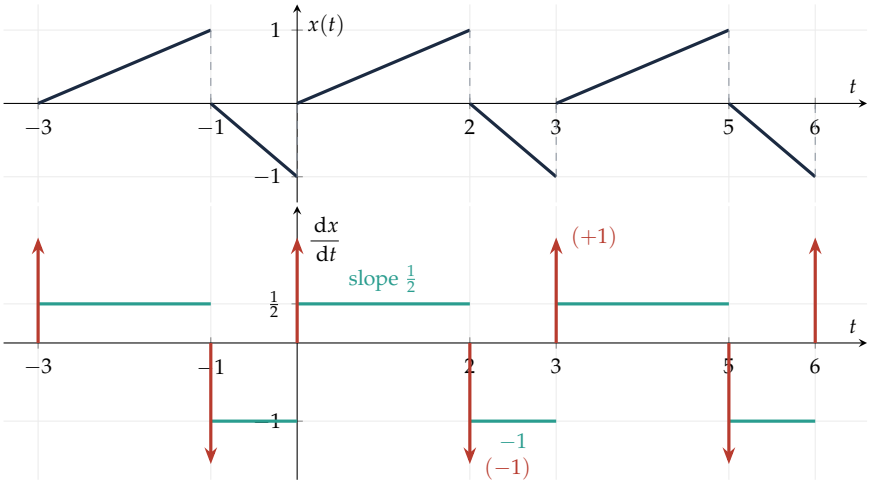
Step 1 — Understand $x(t)$**.** On $[0, 2)$ it rises from 0 to (just below) 1 with slope $\frac{1}{2}$. At $t = 2$ it restarts at $2 - 2 = 0$ and falls with slope -1 to (just above) -1 at $t \to 3$. The next period restarts at $x(3) = 0$. So there are two jumps per period:

$$\begin{aligned} \text{at } t = 2: &\quad x(2^-) = 1 \;\to\; x(2^+) = 0 && (\text{jump } -1), \\ \text{at } t = 3\ (\equiv 0): &\quad x(3^-) = -1 \;\to\; x(3^+) = 0 && (\text{jump } +1). \end{aligned}$$

Step 2 — Differentiate. Smooth pieces give the slopes; each jump of size J gives an impulse $J\,\delta(t - t_0)$:

$$\frac{dx}{dt} = \begin{cases} \frac{1}{2}, & 0 < t < 2 \\ -1, & 2 < t < 3 \end{cases} \; - \delta(t-2) + \delta(t-3), \qquad \text{repeated every 3.}$$

Over the whole axis: $\frac{dx}{dt} = \sum_m \big[\delta(t - 3m) - \delta(t - 2 - 3m)\big]$ + the piecewise-constant slope.



Red arrows: impulses of area $+1$ at $t = 0, \pm 3, 6, \ldots$ and area -1 at $t = 2, 5, -1, \ldots$ (arrow length shows the area, not a height).

Check. The derivative of a periodic signal has zero average: per period $\frac{1}{2}(2) + (-1)(1) + 1 - 1 = 0$ ✓. (Equivalently, its Fourier coefficients are $jk\omega_0 a_k$, which vanish at $k = 0$.)

Answer. $\frac{dx}{dt} = \frac{1}{2}$ on $(0, 2)$, -1 on $(2, 3)$, plus impulses $-\delta(t-2)$ and $+\delta(t-3)$, repeated with period 3.

08

The Fourier Transform: Impulses, Pulses and Applications

QUESTION 8 • 35 MARKS

Q8(a) Transform of shifted impulses MARKS 5 + 5

(i) Using the sifting property of the impulse, find the Fourier transform of $x(t) = \delta(t-a)$.
(ii) The signal $p_k(t)$ consists of k equally spaced unit impulses at $t = 1, 2, \ldots, k$:

$$p_k(t) = \delta(t-1) + \delta(t-2) + \cdots + \delta(t-k) = \sum_{m=1}^{k} \delta(t-m).$$

Using (i), the additivity of the Fourier transform and the sum of a geometric series, show that $P_k(j\omega) = \dfrac{e^{-j\omega}\left(1-e^{-jk\omega}\right)}{1-e^{-j\omega}}$.

SOLUTION

(i) Sifting: $\int f(t)\delta(t-a)\,\mathrm{d}t = f(a)$. With $f(t) = e^{-j\omega t}$:

$$X(j\omega) = \int_{-\infty}^{\infty} \delta(t-a)\, e^{-j\omega t}\,\mathrm{d}t = e^{-j\omega a}.$$

Magnitude 1 at all frequencies (an impulse contains every frequency equally); phase $-\omega a$, a straight line whose slope is the delay.

(ii) Step 1 — additivity. Transform term by term using (i) with $a = m$:

$$P_k(j\omega) = \sum_{m=1}^{k} e^{-j\omega m}.$$

Step 2 — geometric series. Let $q = e^{-j\omega}$ (assume $q \neq 1$, i.e. $\omega \neq 2\pi n$). The sum is $q + q^2 + \cdots + q^k$, first term q, ratio q, k terms:

$$P_k(j\omega) = q\,\frac{1-q^k}{1-q} = \boxed{\frac{e^{-j\omega}\left(1-e^{-jk\omega}\right)}{1-e^{-j\omega}}}$$

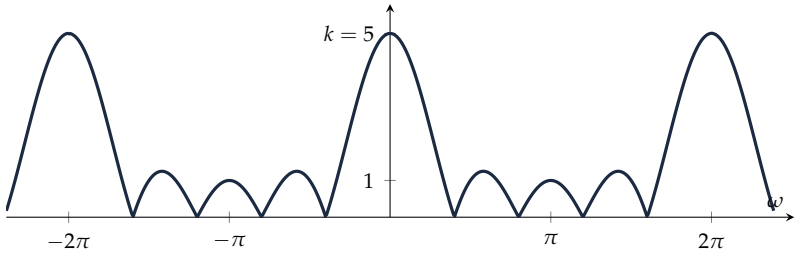
At $\omega = 2\pi n$ every term equals 1 and $P_k = k$ (the limit of the formula).

Step 3 — a useful closed form. Factor half-angles: $1 - e^{-jk\omega} = e^{-jk\omega/2}\, 2j \sin\frac{k\omega}{2}$ and $1 - e^{-j\omega} = e^{-j\omega/2}\, 2j \sin\frac{\omega}{2}$:

$$P_k(j\omega) = e^{-j\omega(k+1)/2}\,\frac{\sin(k\omega/2)}{\sin(\omega/2)}.$$

The phase is linear (centre of the train at $t = \frac{k+1}{2}$) and the magnitude is the periodic "Dirichlet" pattern with peaks k at $\omega = 2\pi n$.

$$|P_k(j\omega)| = |\sin(k\omega/2)/\sin(\omega/2)| \text{ for } k = 5$$



Answer. (i) $\delta(t-a) \longleftrightarrow e^{-j\omega a}$. (ii) $P_k(j\omega) = \sum_{m=1}^{k} e^{-j\omega m} = \dfrac{e^{-j\omega}(1-e^{-jk\omega})}{1-e^{-j\omega}} = e^{-j\omega(k+1)/2}\dfrac{\sin(k\omega/2)}{\sin(\omega/2)}$.

Q8(b) A stepped pulse from rectangular pulses **MARKS 10**

Consider $x(t) = 1$ for $1 < t < 2$, 2 for $2 < t < 3$, 1 for $3 < t < 4$, and 0 otherwise. Given the pair

$$r_T(t) = \begin{cases} 1, & |t| < \frac{T}{2} \\ 0, & \text{otherwise} \end{cases} \qquad \longleftrightarrow \qquad R_T(j\omega) = \frac{2\sin(\omega T/2)}{\omega},$$

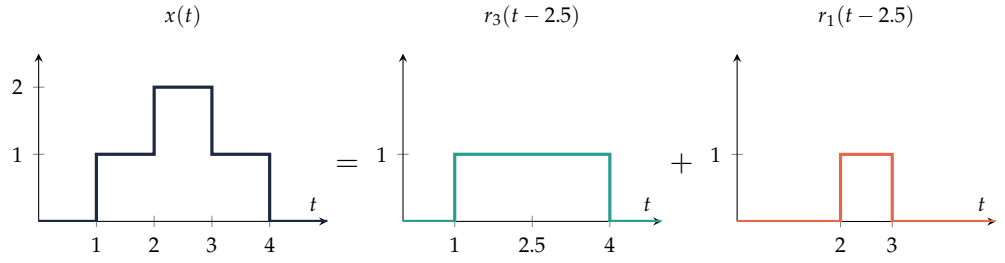
find $X(j\omega)$ without directly evaluating the Fourier transform integral.

SOLUTION

Step 1 — Build x from rectangles. The signal is symmetric about $t = 2.5$. It is a wide pulse of height 1 on $(1,4)$ plus a narrow pulse of height 1 on $(2,3)$:

$$x(t) = r_3(t-2.5) + r_1(t-2.5).$$

(Check: on $(2,3)$ both are 1, total 2; on $(1,2)$ and $(3,4)$ only the wide one, total 1.)



Step 2 — Transform each piece. From the given pair, $r_3 \longleftrightarrow \dfrac{2\sin(3\omega/2)}{\omega}$ and $r_1 \longleftrightarrow \dfrac{2\sin(\omega/2)}{\omega}$. A shift by 2.5 multiplies by $e^{-j2.5\omega}$ (part (a)(i): shifting is convolution with $\delta(t-2.5)$).

Step 3 — Linearity.

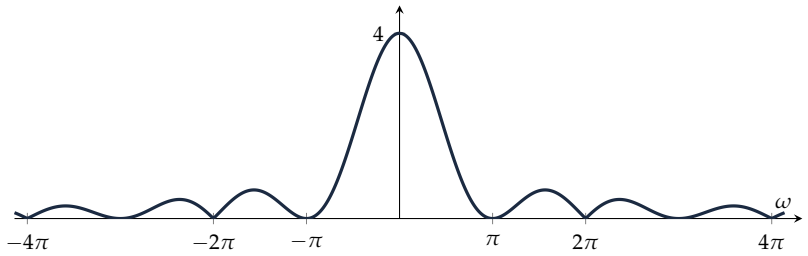
$$\boxed{X(j\omega) = e^{-j2.5\omega}\,\frac{2\big[\sin(1.5\,\omega) + \sin(0.5\,\omega)\big]}{\omega}}$$

Using $\sin A + \sin B = 2\sin\frac{A+B}{2}\cos\frac{A-B}{2}$ (with $A = 1.5\omega$, $B = 0.5\omega$), an equivalent form is

$$X(j\omega) = e^{-j2.5\omega}\,\frac{4\sin\omega\,\cos(\omega/2)}{\omega}.$$

Check. $X(0) = \lim \frac{2(1.5\omega+0.5\omega)}{\omega} = 4 = \int x(t)\,dt = 1+2+1$ ✓. The magnitude is real and even (the pulse is symmetric about its centre); the shift contributes only the linear phase -2.5ω.

$$|X(j\omega)| = |2[\sin 1.5\omega + \sin 0.5\omega]/\omega|$$



Answer. $X(j\omega) = e^{-j2.5\omega}\left[\frac{2\sin(3\omega/2)}{\omega} + \frac{2\sin(\omega/2)}{\omega}\right] = e^{-j2.5\omega}\frac{4\sin\omega\cos(\omega/2)}{\omega}$, with $X(0) = 4$.

Q8(c) Three applications of the Fourier transform

MARKS 5 + 5 + 5

Briefly describe how the Fourier transform can be used in each scenario. (i) An image of a metal surface contains cracks and object boundaries; we need an image in which these edges are emphasised. (ii) A large image must be transmitted through a slow link; we have to compress it while retaining its main visual appearance. (iii) A music recording is corrupted by persistent noise at frequencies above the useful range of the music; we have to reduce this noise without disrupting the music.

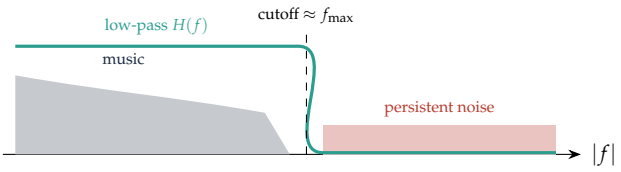
SOLUTION

KEY IDEA Smooth regions = low frequencies; sharp changes (edges, fine detail, hiss) = high frequencies. Filtering in the Fourier domain is just multiplying the spectrum by a mask H and transforming back, since $y = x * h \iff Y = XH$.

(i) Edge enhancement — high-pass filtering. Take the 2-D Fourier transform $X(u,v)$ of the image. Cracks and boundaries are abrupt changes in brightness, so their energy sits at *high spatial frequencies*, while the smooth background is near $(0,0)$. Multiply by a high-pass mask $H(u,v)$ that suppresses frequencies near the origin and keeps (or boosts) the outer region, then take the inverse transform. The result shows the edges bright on a dark background; adding it back to the original ("high-boost"/sharpening) gives an image with emphasised edges. (Equivalently, differentiation $\leftrightarrow$ multiplication by $j\omega$, which grows with frequency.)

(ii) Compression — keep the strong low-frequency coefficients. For natural images most of the energy is concentrated in a small number of low-frequency coefficients (energy compaction; Parseval says energy is preserved, so we can see where it is). Transform the image (in practice blockwise, as with the DCT in JPEG), *quantise* the coefficients and *discard* the many small high-frequency ones, then send only the remaining few (entropy-coded). The receiver inverse-transforms them: the main shapes and shading survive; only fine detail is lost, which the eye hardly notices.

(iii) Noise removal — low-pass filtering. Compute the spectrum of the recording; the music occupies $|f| \le f_{\max}$ and the persistent noise lies above it. Apply a low-pass filter with cutoff just above $f_{\max}$ (multiply the spectrum by $H(f) = 1$ for $|f| \le f_{\max}$, ≈ 0 above, with a smooth transition and linear phase so the waveform is not distorted), then inverse transform (or use the equivalent FIR filter $h = \mathcal{F}^{-1}\{H\}$ in real time). The music band is untouched while the out-of-band noise is removed.



Answer. (i) High-pass filter the 2-D spectrum (edges are high-frequency) and add back to sharpen; (ii) transform, keep and quantise the few large low-frequency coefficients, drop the small high-frequency ones (JPEG-style); (iii) low-pass filter with cutoff just above the music band.